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PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Air and Water Resource Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
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PI Email:	cmitchel@purdue.edu	**Fax:**	FY 765-494-0391
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-1347
Organization Name:	Purdue University		
PI Address 1:	Department of Horticulture and Landscape Architecture		
PI Address 2:	625 Agriculture Mall Dr		
PI Web Page:	http://www.alsnscort.org		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Banks, Katherine (Purdue University) Jones, Kimberly (Howard University) Alleman, James (Purdue University) Blatchley, Ernest (Purdue University) Heber, Al (Purdue University)		
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Task Description:

Providing clean water and air for long-term exploration missions is a top priority for the NSCORT research team. The Air and Water Research Group is focusing on development of innovative technology for wastewater treatment and recycling, drinking water treatment and safety, and removal of trace contaminants from air. For BREATHe: • Prepare a conceptual design of a dual treatment process (liquid and gaseous effluents) in a trickling biofilter reactor. • Construct and operate a representative pilot-scale bio-trickling filter treatment system. • Develop and experimentally verify a mathematical model for the BREATHe system. • Identify and optimize system design parameters, such as gas/liquid flow rate, to maximize treatment efficiency.
For Membrane Filtration Processes: The goal of recent work in membrane processes for NSCORT involves quantifying fouling mechanisms and rejection properties of both membrane systems in an effort to link physical chemical characteristics of the membranes to solute attachment on the membrane surface and solute rejection. Objectives of this project are to • Evaluate suite of membranes for flux and contaminant rejection. • Quantify fouling mechanisms in MF/UF system. • Increase ion rejection for RO/NF system.

For Iodine Chemical Water Disinfection: Four research goals for investigating UV and iodine disinfection are listed below. • Determine whether complementary disinfection using UV radiation and iodine will provide more effective microbial inactivation than either method alone. • Develop a chemical actinometer that can monitor the UV disinfection process, in addition to providing sufficient quantities of iodine to achieve residual disinfection. The dual purpose of the chemical actinometer may reduce chemical restock requirements. • Investigate whether the microbial check valve (MCV), which was developed in the 1970s by NASA, can be used to store and deliver the photochemically-produced iodine. • Evaluate the effectiveness of using Vitamin C to remove iodine from potable water, while providing a beneficial residual for the astronauts.

Water Disinfection UV Irradiation (Physical): ESM calculations for the Mars surface mission (duration 600 days) were performed for the proposed UV disinfection reactor, resulting in an overall ESM estimate of 117 kg. Figure 1 is s schematic representation of the UV disinfection reactor geometry that the ESM values were based upon. Figure¬ 2 represents ESM partitions based on individual components. The greatest contribution to the ESM value comes from the power consumption by the UV lamp. It should be noted that alternative reactor designs will be investigated as part of this project that are likely to alter the ESM estimate described above.

The greatest contribution to the ESM value comes from the power consumption by the UV lamp. It is expected that power consumption can be reduced by increasing the efficiency of the UV lamp. Crew time can be decreased by reducing the maintenance requirement, and mass reduction can be achieved if the reactor is constructed out of a material lighter than stainless steel. All these possibilities are being investigated as part of the alternative reactor design mentioned above

For Gas Rivitalization using Biofiltration: Our goal is to develop bio-regenerative air cleaning systems to efficiently remove the gas phase contaminants that will be generated in the cabin during long term space missions. The Bio-Regenerative Environmental Treatment for Health (BREATHe) has been proposed as an alternate system to decontaminate cabin air using biofiltration. Our ultimate research goal is to optimize design and operation of air biofiltration in the ALS indoor air system. The goal is to minimize the ESM burden and to maximize biofilter performance simultaneously. The ESM reduction may be achieved by optimizing gas/water loading and biofilter depth, and selecting biofilter media with a large surface area. Our approach is to optimize design and operation based on theoretical modeling first, and then to verify model results by experimentation. Specific goals are as follows. 1. Simulation of biofiltration and indoor air quality in the ALS system • Develop mathematical models to study the effects of biofilter design and operation on air cleaning performance. • Simulate the airflow and the indoor air quality in an ALS cabin using computational fluid dynamics and mass transfer programs and predict air purification by biofiltration. • Apply simulation results to design and control of experimental and full-scale biofilters.

2. Experimental setup of the BREATHe II system • Compare results with biofiltration process modeling. • Experimentally observe effects of media depth, media type, microbial growth, media clogging, gas residence time, and using of condensate on biofiltration performance. • Evaluate microbial degradation of several gaseous contaminants.

For LiFT: • The primary goal is potable water extraction from urine; however, additional waste streams will be tested based on the same principle technology. • Research geared towards reduction of ESM compared to existing water recovery technologies. i.e. VCD and VPCAR

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Development of new water and air-treatment technologies for environmental protection
Will impact air-treatment technology and air quality control strategies used in residential and commercial buildings and in commercial aircraft.

Further development of fundamental photochemical reactor theory as it relates to systems used for disinfection of water and air.

Potable Water Disinfection Subject to Extended Space Travel Constraints. A numerical method for prediction of UV disinfection process efficacy has been developed and experimentally tested. Experimental results indicate that the model provides a reasonably accurate representation of disinfection efficacy in the UV reactor system, and the fundamental physical behavior that governs performance. Analytical methods for the detection of various iodine species have been identified. Our developed iodine speciation model allows the concentration of various iodine species to be determined using information such as pH and total iodine concentration.
Liquid Freeze-Thaw (LiFT) Urine and RO Brine Processing for Advanced Water Recovery and Salt Separation. Collaborations with a freeze concentration company have resulted in a planned pilot plant study of water extraction recovery rates and cleaning capabilities of a model urine solution. The study will consist of two phases. Phase I involves off-site testing located at the company's pilot plant. Upon completion and process verification, Phase II will include on-site testing of urine samples at Purdue using the company's freeze concentration unit.

Task Progress:	Membrane Processes in Advanced Live Support. Ion rejection was evaluated from modified and unmodified membranes. Increased electrostatic interactions near the membrane surface increased ion rejection from these membranes. Low salt flux indicates high ion rejection. The next goal is to verify these results for other membranes and determine if NF membranes can compete with RO membranes for ion rejection. Positive results would indicate that NF membranes can be modified to have high monovalent ion rejection, but with much lower operating pressures. Bio-Regenerative Environmental Air Treatment for Health (BREATHe): Integrated STAR Off-Gas, Cabin Air, and Grey Water Processing. Respirometry has been successfully used to provide general information about the biodegradability of target surfactants. Carbon dioxide production was used as a measure of ultimate biodegradation and both surfactants were found to be readily biodegradable. A test run of the proto-type BREATHe reactors was successful. Through mathematical modeling, we found that surface area of packing material, gas velocity, and liquid velocity were the most sensitive experimental parameters. Gas Phase Revitalization Using Biofilters in ALS. Programming all simulation models for predicting biofiltration performance and ALS indoor air quality has been completed. Although the general framework for modeling the biofiltration process has been completed, the chemical kinetics and equilibrium equations for microbial degradation will be added to the models after initial experiments. The indoor air quality simulations are being conducted by commercial programs for computational fluid dynamics. In addition, construction of 24 biofilters is complete. Trainees from our research programs included nine graduate students and five undergraduate students. Our NSCORT group interacts often through teleconferencing meetings. Collaboration with NASA researchers has occurred in several of the research projects. Significant progress is being made toward our objective of optimizing treatment efficiency while significantly reducing ESM.
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